Topic 6: Bandwidth Utilization

CSE 4405: Data and Telecommunications

Department of CSE

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Bandwidth Utilization

Definition

Bandwidth utilization is the wise use of available bandwidth to achieve communication goals.

- **Efficiency goal:** serve more traffic on the same infrastructure.
- **Security/robustness goal:** reduce eavesdropping and jamming impact.
- In this chapter, these goals are addressed through **multiplexing** and **spread spectrum**.

6.1 Multiplexing: Scope

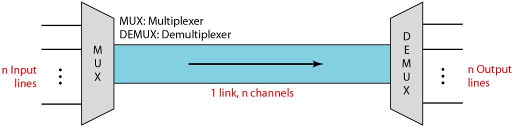
- Multiplexing allows simultaneous transmission of multiple signals over one shared link.
- It is used when link bandwidth is larger than each individual source requirement.
- As communication demand grows, multiplexing becomes essential for cost-effective scaling.

Topics in this section

- Frequency-Division Multiplexing (FDM)
- Wavelength-Division Multiplexing (WDM)
- Synchronous Time-Division Multiplexing (TDM)
- Statistical Time-Division Multiplexing

Multiplexing Basics: Link vs Channel

- **Link**: the physical path (cable, fiber, radio path).
- **Channel**: a logical portion of the link carrying one transmission stream.
- Multiplexing divides one physical link into multiple channels.



Multiplexing and Demultiplexing

Multiplexing

Combine several input streams into one composite stream (*many-to-one*).

Demultiplexing

Split the composite stream back into its component streams (*one-to-many*) and deliver each to the intended receiver.

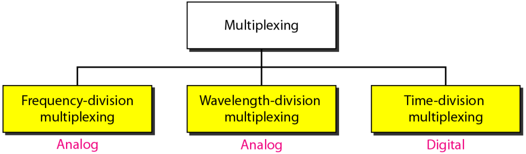


Figure: The shared link carries a combined signal between a MUX and a DEMUX.

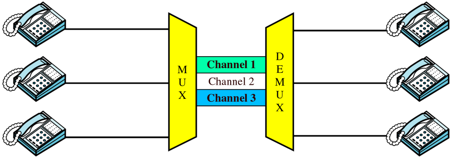
FDM: Core Idea

Frequency-Division Multiplexing (FDM)

FDM is an **analog multiplexing** technique that combines analog signals by assigning each one a separate carrier frequency band.

User 1 voice: 0–4 kHz → shifted to 20–24 kHz (carrier @22Hz)
User 2 voice: 0–4 kHz → shifted to 30–34 kHz (carrier @32Hz)
User 3 voice: 0–4 kHz → shifted to 40–44 kHz (carrier @42Hz)

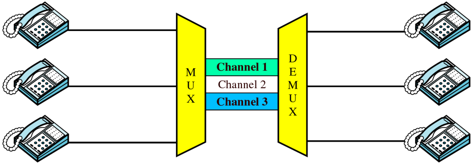
- Each source occupies a different frequency slot.
- All slots are transmitted simultaneously over the same link.
- Receiver-side filters separate the bands before demodulation.



FDM Process: Multiplexing Side

- Each telephone/data source generates a baseband signal.
- Each signal modulates a different carrier ($f_1, f_2, f_3, \ldots$).
- Modulated signals are combined into one composite spectrum.

Result: one shared analog link carries multiple conversations simultaneously.



FDM Multiplexing in the Frequency Domain

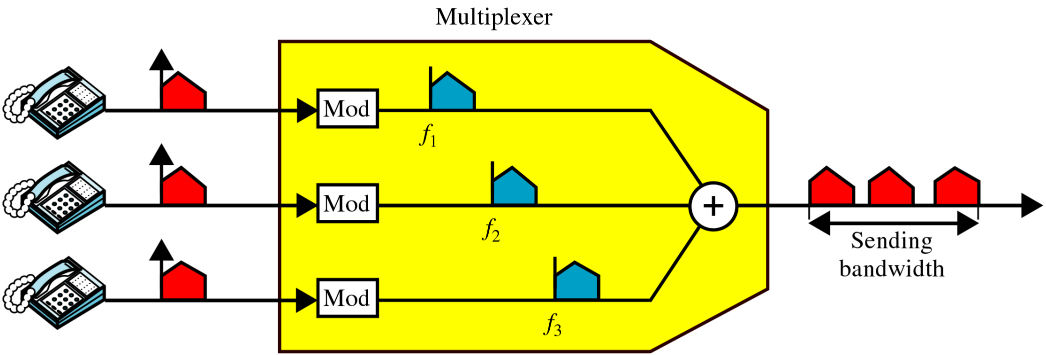
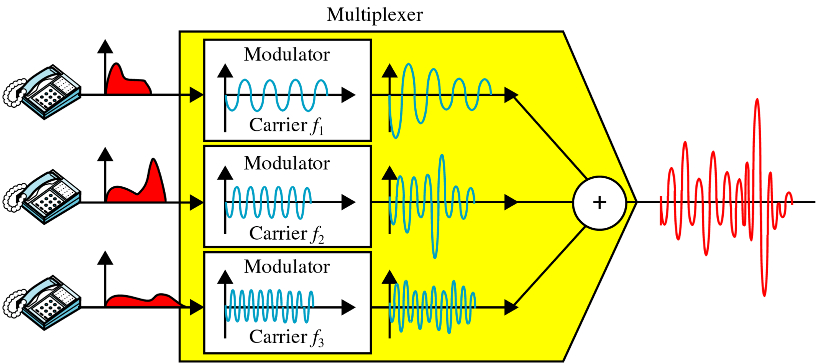


Figure: Separate subbands are allocated side-by-side across frequency, then combined onto one link.

FDM Multiplexing in the Time Domain

Figure: Although users share one medium, each keeps a dedicated frequency band continuously over time.



FDM Process: Demultiplexing Side

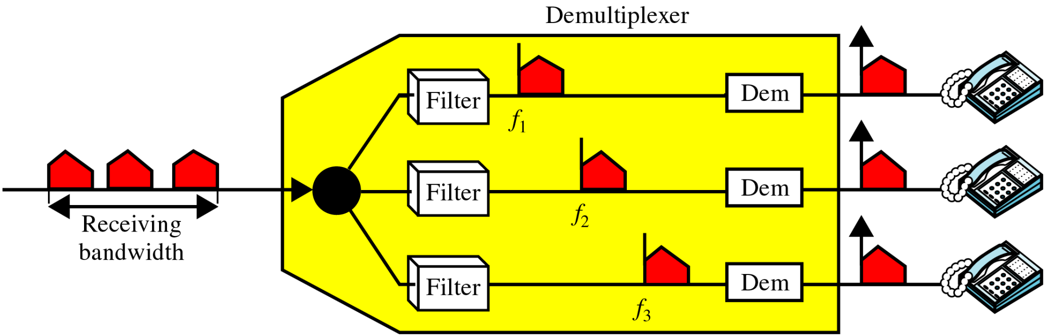
- At the receiver, the composite signal enters a demultiplexer.
- Band-pass filters isolate each carrier band.
- Each isolated band is demodulated to recover the original signal.
- Recovered streams are forwarded to the corresponding destinations.

Key point

Demultiplexing reverses multiplexing by frequency-selective separation.

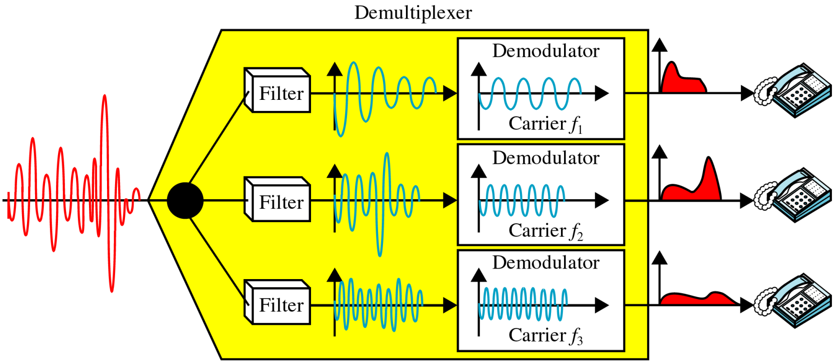
FDM Demultiplexing in the Frequency Domain

Figure: The composite spectrum is split back into original channel bands before delivery.



FDM Demultiplexing in the Time Domain

Figure: Each destination receives only its own continuous signal after filtering and demodulation.



Example 6.1: Three Voice Channels

Problem

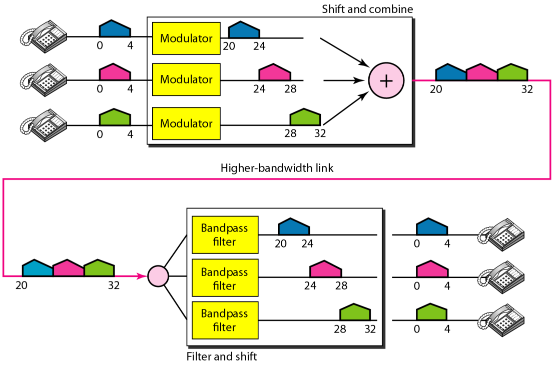
- One voice channel bandwidth: 4 kHz
- Combine 3 channels into 20 to 32 kHz (12 kHz total)
- Assume no guard bands

Solution idea

- Shift channels to 20–24, 24–28, and 28–32 kHz
- Then combine the three modulated channels into one link spectrum

Figure 6.6: Example 6.1

Figure: Three 4 kHz channels are translated to adjacent 4 kHz bands and combined.



Example 6.2: Guard Bands

Problem

- 5 channels, each 100 kHz
- Guard band required between adjacent channels: 10 kHz

Minimum link bandwidth

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz}$$

Why 4 guard bands? For N channels in one block, there are $N - 1$ boundaries between adjacent channels.

Figure 6.7: Example 6.2

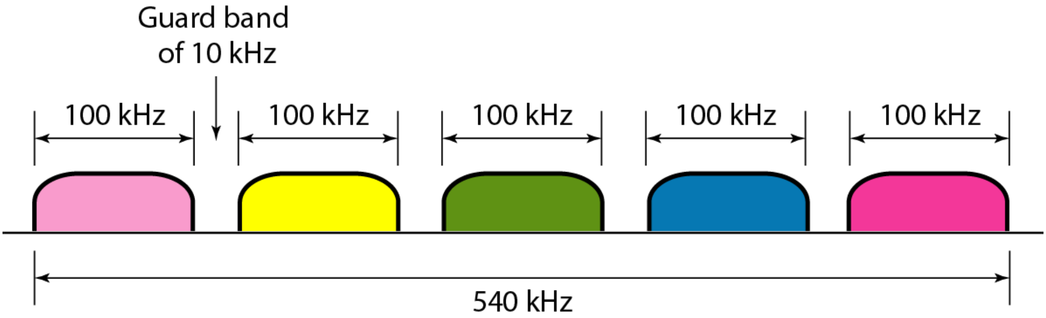


Figure: Guard bands are inserted between channel bands to reduce adjacent-channel interference.

Example 6.3: Digital Sources over an Analog Satellite Channel

Problem

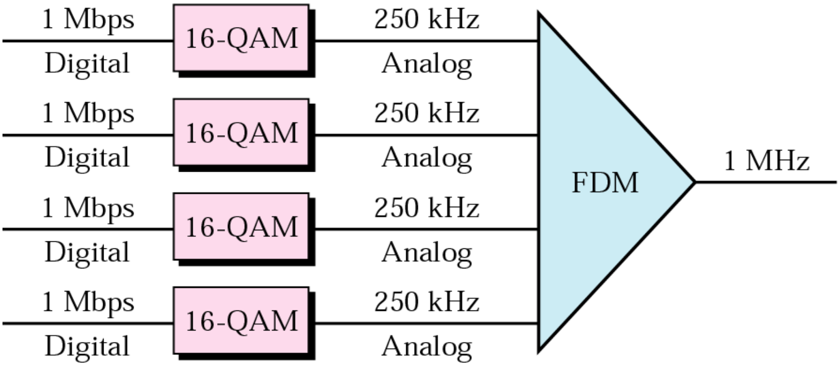
- Four digital channels, each 1 Mbps
- One analog satellite channel, total bandwidth 1 MHz

Design idea

- Split satellite bandwidth into four 250 kHz subchannels
- Modulate each digital source to fit its subchannel
- One possible modulation choice: 16-QAM (4 bits/symbol)

Figure 6.8: Example 6.3

Figure: Digital streams are modulated, frequency-separated, and carried together over one analog satellite band.



FDM Application: Cable Television

- Typical coaxial cable bandwidth: approximately 500 MHz
- One TV channel needs about 6 MHz
- Theoretical capacity: $\lfloor 500/6 \rfloor \approx 83$ channels

Engineering note

Practical channel count is lower after accounting for guard bands, control channels, and spectrum planning constraints.

```
A control channel is an actual channel used to carry system/signaling information, not
user voice/data.For example, in AMPS/mobile systems, control channels are used for things
like:
finding the network
setting up calls
paging a phone
assigning a voice channel
handoff coordination
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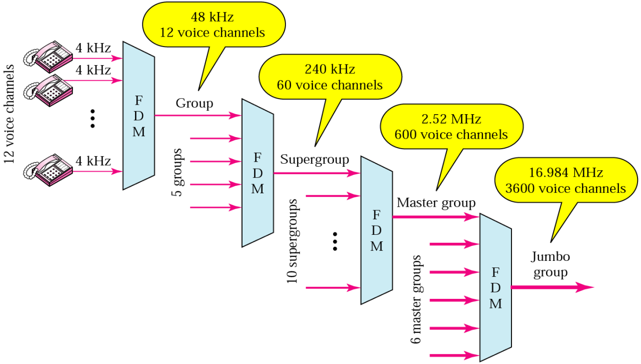
Analog Hierarchy in Telephony

Figure: Telephone infrastructure multiplexing hierarchy



Example 6.4: AMPS Capacity

Given

- Uplink band: 824–849 MHz, downlink band: 869–894 MHz
- Per-user bandwidth per direction: 30 kHz

Capacity estimate

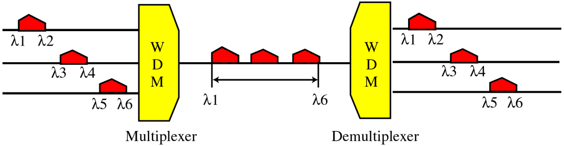
- Bandwidth per band: 25 MHz $\Rightarrow 25,000/30 \approx 833$ channels
- Practical allocation used 832 channels
- 42 channels reserved for control $\Rightarrow$ 790 user channels

WDM: Optical Counterpart of FDM

WDM idea

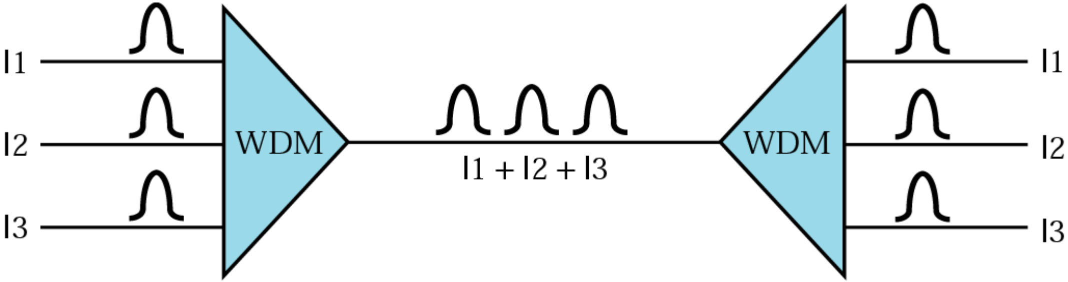
WDM is conceptually the same as FDM, but in fiber optics: channels are separated by **wavelength** instead of radio frequency bands.

- Multiple optical carriers ($\lambda_1, \lambda_2, \lambda_3, \ldots$) share one fiber.
- Optical MUX combines wavelengths; optical DEMUX separates them.
- Used to exploit the very high bandwidth of fiber links.



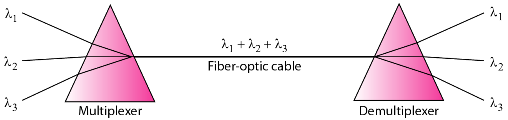
WDM in One Picture

Figure: Different light wavelengths coexist on the same fiber, each carrying an independent data channel.



Prism Intuition for WDM

- A prism bends light by wavelength-dependent refraction.
- The same physical principle explains optical combining and splitting.
- Optical transport systems such as SONET/SDH can use WDM layers for capacity growth.



TDM: Core Idea

Time-Division Multiplexing (TDM)

TDM is a **digital multiplexing** process in which multiple low-rate channels share one high-rate link by taking turns in time.

- Use TDM when link capacity exceeds per-source data rate.
- Time is divided into slots; slots are grouped into frames.
- Each source transmits in its assigned slot(s).

- **Synchronous TDM**: fixed slot assignment in every frame.
- **Statistical TDM**: slots assigned dynamically to active inputs.

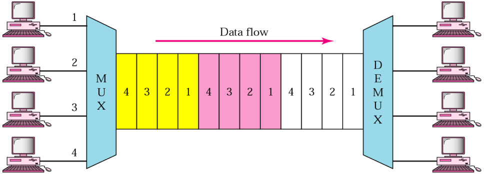
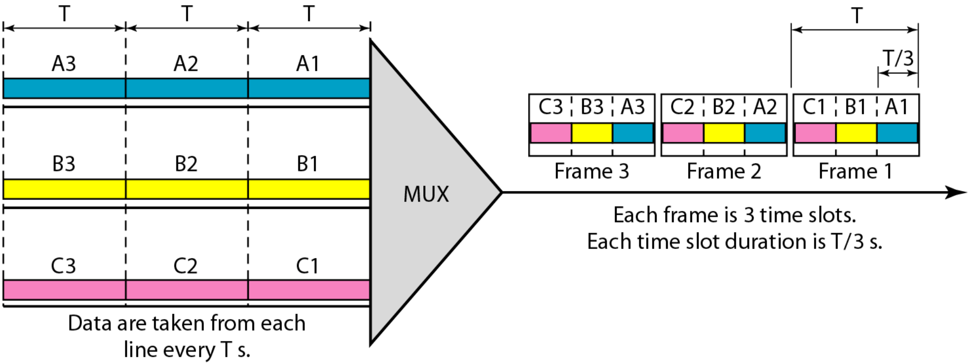


Figure: Both schemes share a high-bandwidth link in time; they differ in slot assignment policy.

--> Synchronous TDM: every input gets a fixed time slot in every frame, even if it has no data to send. The advantage is that it is simple. The demux knows:
first slot → User 1
second slot → User 2
third slot → User 3
fourth slot → User 4
The disadvantage is wasted space. If User 3 has no data, User 3's slot may still be reserved and go empty.
In statistical TDM, slots are given only to active users. So if only users 1 and 4 currently have data, the frame might contain only:
| 1 | 4 | 1 | 4 |
If later users 2 and 3 become active, the slots can change dynamically:
| 2 | 1 | 3 | 4 |
The advantage is better efficiency because empty slots are not wasted.
The disadvantage is that the receiver needs extra information to know which slot belongs to which user. In synchronous TDM, position tells you the user. In statistical TDM, because the order can change, each slot usually needs an address/header.

Figure 6.13: Synchronous TDM

Figure: In synchronous TDM with n inputs, link rate is n times faster and each output slot is $1/n$ of input-unit duration.



Example 6.5: Timing from Figure 6.13

Given each input rate is 1 kbps, and one bit is multiplexed per slot.

1. Input bit duration $= 1/1000$ s $= 1$ ms
2. Output slot duration $= 1/3$ ms (three inputs)
3. Frame duration $= 3 \times (1/3 \text{ ms}) = 1$ ms

Takeaway: frame duration equals the input unit duration when one unit per source is interleaved per frame.

Example 6.6: Bit and Frame Rates

Given Figure 6.14 has four 1 Mbps inputs, one-bit unit.

1. Input bit duration: $1/1$ Mbps $= 1\ \mu s$
2. Output bit duration: $1/4\ \mu s$
3. Output bit rate: 4 Mbps
4. Output frame rate: $1,000,000$ frames/s

Check: frame rate $\times$ bits/frame $= 10^6 \times 4 = 4 \times 10^6$ bps.

Figure 6.14: Synchronous TDM (Example 6.6)

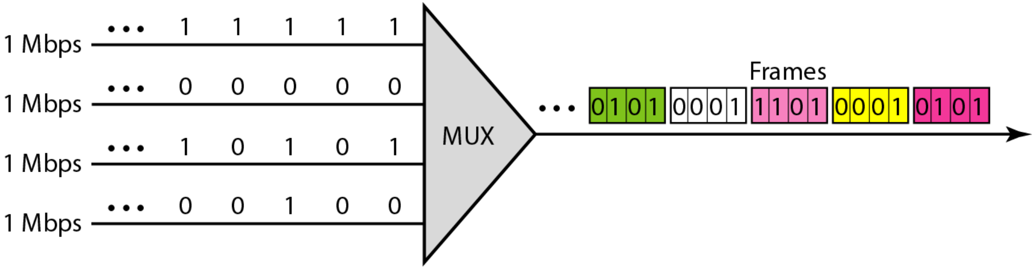


Figure: One bit from each input is interleaved into each frame, creating a higher-rate output stream.

Example 6.7: Four 1 kbps Inputs

Given four 1 kbps connections, unit = 1 bit.

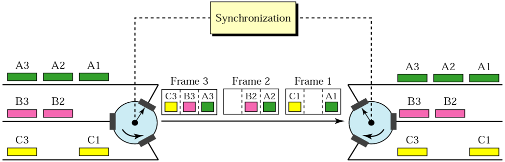
1. Input bit duration: $1/1$ kbps $= 1$ ms
2. Link rate: 4×1 kbps $= 4$ kbps
3. Slot duration: $1/4$ ms $= 250\ \mu s$
4. Frame duration: $4 \times 250\ \mu s = 1$ ms

Synchronous TDM: Interleaving

- Interleaving can be visualized as a very fast rotating switch.
- On the MUX side, each input sends when its slot position arrives.
- On the DEMUX side, the receiving switch distributes units in the same order.
- Correct operation requires synchronized slot timing.

Synchronous TDM: Empty Slots

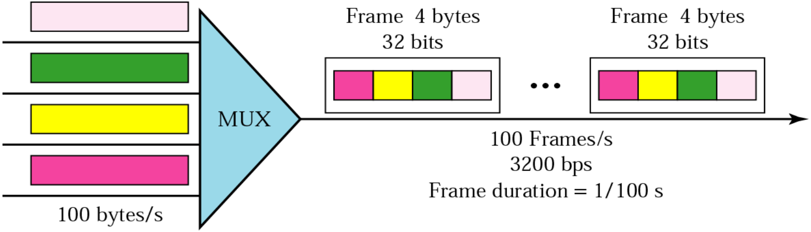
- Fixed slot assignment is simple and predictable.
- If a source is idle, its slot in that frame becomes empty.
- Empty slots reduce bandwidth efficiency for bursty traffic.



Example 6.8: Byte-Level Multiplexing

Given 4 channels, each 100 bytes/s, multiplex 1 byte per channel per frame.

- Frame size: 4×1 byte $= 4$ bytes $= 32$ bits
- Frame rate: 100 frames/s (to serve each channel's 100 bytes/s)
- Frame duration: $1/100 = 0.01$ s $= 10$ ms
- Link bit rate: $100 \times 32 = 3200$ bps $= 3.2$ kbps



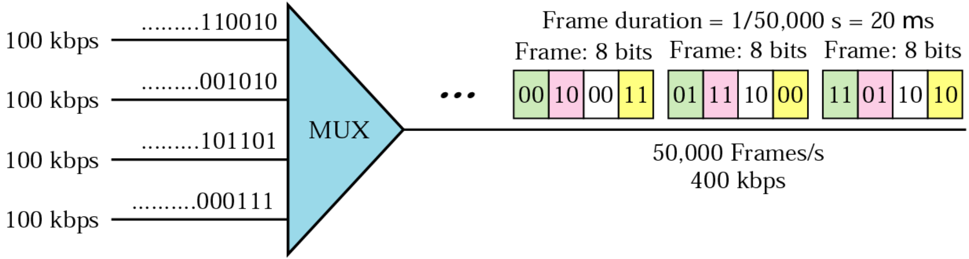
Example 6.9: 2-Bit Slots

Given four 100 kbps channels, time slot $= 2$ bits/channel.

- Frame carries $4 \times 2 = 8$ bits.
- Frame rate: $100{,}000/2 = 50{,}000$ frames/s
- Frame duration: $1/50{,}000 = 20\ \mu s$
- Link bit rate: $50{,}000 \times 8 = 400$ kbps
- Bit duration on link: $1/400{,}000 = 2.5\ \mu s$

Example 6.9 Output Illustration

Figure: Arbitrary 2-bit chunks from each channel are interleaved into 8-bit frames at 50,000 frames/s.

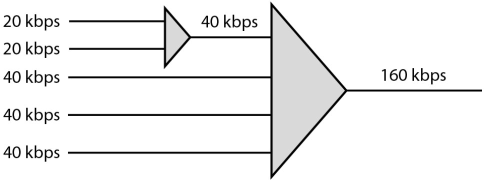


Synchronous TDM: Data-Rate Management

- Real systems often have unequal input rates.
- Three standard strategies are used to handle rate disparity:
  - Multilevel multiplexing
  - Multiple-slot allocation
  - Pulse stuffing (bit padding)

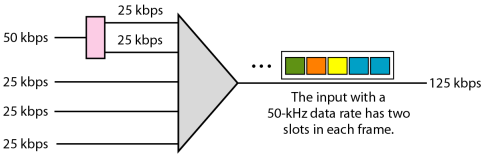
Multilevel Multiplexing

- Used when one input rate is an integer multiple of another.
- First, combine lower-rate inputs into an intermediate stream.
- Then perform higher-level multiplexing with other streams.
- This creates a clean hierarchical design.



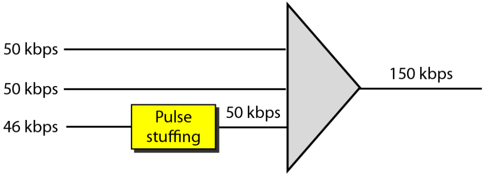
Multiple-Slot Allocation

- Give faster sources more than one slot per frame.
- A serial-to-parallel converter can split one high-rate input into multiple logical slot streams.
- Example: a 50 kbps line may be assigned two slots while lower-rate lines get one.



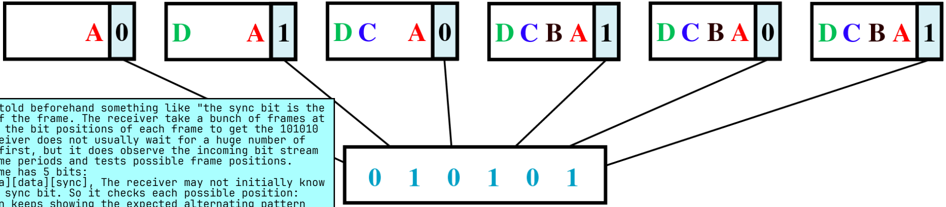
Pulse Stuffing (Bit Padding)

- Used when rates are not integer multiples.
- Choose the highest source rate as the reference.
- Add dummy bits to lower-rate streams to match timing.
- Also called **bit padding** or **bit stuffing**.



Framing Synchronization

- If MUX and DEMUX lose frame alignment, bits can be delivered to wrong channels.
- Framing bits are inserted so the receiver can detect slot boundaries.
- A common pattern is a 1-bit sync stream (e.g., alternating 101010 . . .).
- Synchronization bits are typically placed at a fixed frame position.



Synchronization pattern

The receiver is told beforehand something like "the sync bit is the first/last bit of the frame. The receiver take a bunch of frames at once and matches the bit positions of each frame to get the 101010 pattern. The receiver does not usually wait for a huge number of complete frames first, but it does observe the incoming bit stream over several frame periods and tests possible frame positions.
Suppose each frame has 5 bits:
[data][data][data][data][sync], The receiver may not initially know which bit is the sync bit. So it checks each possible position:
When one position keeps showing the expected alternating pattern 101010..., the receiver says:
That position is the sync bit.
Now I know the frame boundary.

Example 6.10: Character Streams with Framing Bit

Given 4 sources, each 250 characters/s, 8 bits/character, plus 1 framing bit/frame.

1. Source rate: $250 \times 8 = 2000$ bps $= 2$ kbps
2. Character duration: $1/250 = 4$ ms
3. Frame rate: 250 frames/s
4. Frame duration: 4 ms
5. Bits/frame: $4 \times 8 + 1 = 33$
6. Link rate: $250 \times 33 = 8250$ bps

Example 6.11: Unequal Input Rates

Problem Multiplex 100 kbps and 200 kbps channels. **Design**

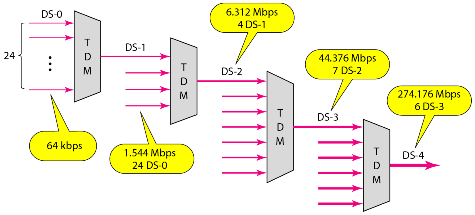
- Allocate 1 slot/frame to 100 kbps channel.
- Allocate 2 slots/frame to 200 kbps channel.
- So each frame carries 3 bits total.

Results

- Frame rate: 100,000 frames/s
- Frame duration: 10 μs
- Link bit rate: $100{,}000 \times 3 = 300$ kbps

Digital Signal (DS) Service

- Digital hierarchy defines standard multiplexed carrier rates.
- Compared with analog trunks, digital services are generally:
  - less sensitive to noise
  - more cost-effective for large-scale deployment



DS Service Rate Family

Level	Rate	Composition intuition
DS-0	64 kbps	Single digital voice/data channel
DS-1	1.544 Mbps	24×64 kbps plus overhead
DS-2	6.312 Mbps	96×64 kbps plus overhead
DS-3	44.376 Mbps	672×64 kbps plus overhead
DS-4	274.176 Mbps	4032×64 kbps plus overhead

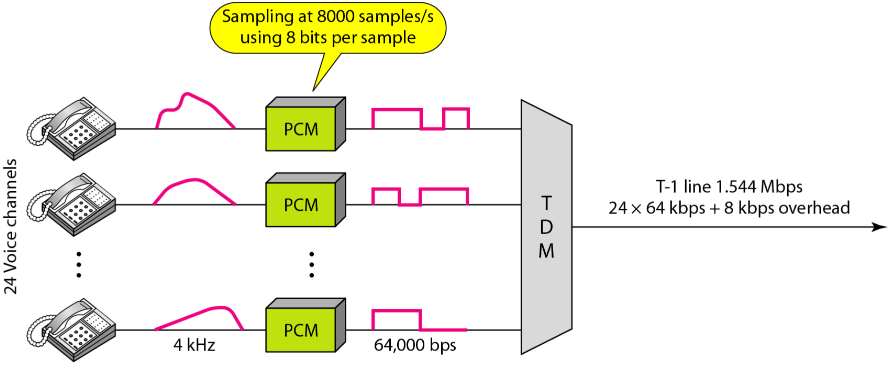
Table 6.1: DS and T-Line Rates

Figure: DS levels map to practical T-carrier transmission rates used in digital telephony infrastructure.

Service	*Line*	*Rate (Mbps)*	*Voice Channels*
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

Figure 6.24: T-1 for Telephone Multiplexing

Figure: A T-1 line multiplexes many voice channels into one framed digital stream for trunk transport.



Frame size

$$193 = 24 \times 8 + 1$$

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data is sent by samples
instead of by bits
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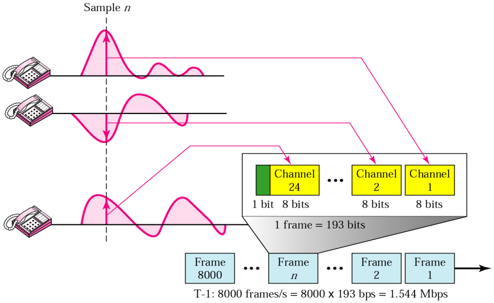
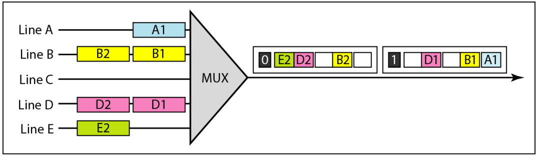
- 24 channels $\times$ 8 bits $=$ 192 payload bits
- 1 additional framing bit for synchronization



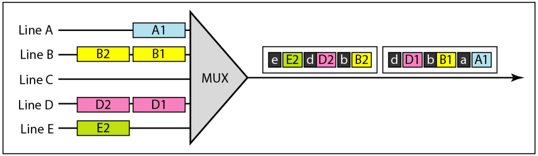
Statistical TDM: Motivation

- Slots are assigned dynamically to active sources.
- In synchronous TDM, fixed slots can be wasted when lines are idle.
- In statistical TDM, no slot is left empty while any input has data waiting.
- This improves bandwidth efficiency for bursty traffic.

Figure 6.26: Slot Comparison



a. Synchronous TDM



b. Statistical TDM

Statistical TDM: Addressing and Synchronization

- Because slot positions vary, each slot carries **address information**.
- If N output lines exist, n address bits are needed where:

$$n = \log_2 N$$

- Frame-wide synchronization bits used in synchronous TDM are typically unnecessary in the same form.
- The receiver uses slot headers to deliver payloads correctly.

6.2 Spread Spectrum: Scope

- Spread spectrum (SS) intentionally uses bandwidth wider than the minimum required for raw data.
- Main goals are **anti-jamming robustness** and **reduced interception vulnerability**.
- Redundancy is introduced during spreading to gain these properties.

Techniques in this section

- Frequency-Hopping Spread Spectrum (FHSS)
- Direct-Sequence Spread Spectrum (DSSS)

Instead of using only the minimum bandwidth needed, we intentionally spread the signal over a much wider frequency range.

1. Anti-jamming robustness
A jammer usually tries to disturb your signal by transmitting noise/interference at the same frequency. If your signal is concentrated in one narrow band, the jammer can easily attack it. This protects against someone intentionally trying to sabotage communication by transmitting interference/noise. That attacker is called a jammer. Without spread spectrum, if your signal uses one narrow frequency band, the jammer can target that band and block the link. With spread spectrum, the signal is spread over many frequencies, so the jammer has a harder job. They would need to jam a much wider band or know the spreading/hopping pattern.
Its about protecting availability — keeping communication working.

2. Reduced interception vulnerability
Because the signal is spread out over a wide frequency range, its power at any one frequency can be very low. To someone who does not know the spreading pattern, it can look like background noise. So spread spectrum can make the signal harder to detect, intercept, or separate from noise. Because the signal is spread over a wide bandwidth, its power at each frequency can be very low. To someone who does not know the spreading code or hopping pattern, it may look like weak noise.
It is about privacy/stealth — making the communication harder to notice or intercept.

What does “redundancy is introduced” mean?
It means the same data is represented using more transmitted signal elements than the minimum needed. For example, instead of sending one data bit directly as one symbol, spread spectrum may represent that bit using a longer code sequence. Example:
Bit 1 → 101101001011
Bit 0 → 010010110100
This is about making it harder for an unwanted listener to detect or eavesdrop on the signal.

Figure 6.27: Spread Spectrum Principles

- Allocate more bandwidth than the original data stream strictly needs.
- Perform spreading after source signal generation.
- Receiver performs inverse operation (despreading) using matching strategy/code.

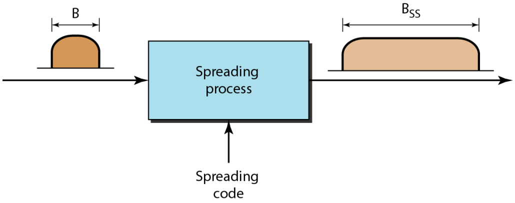
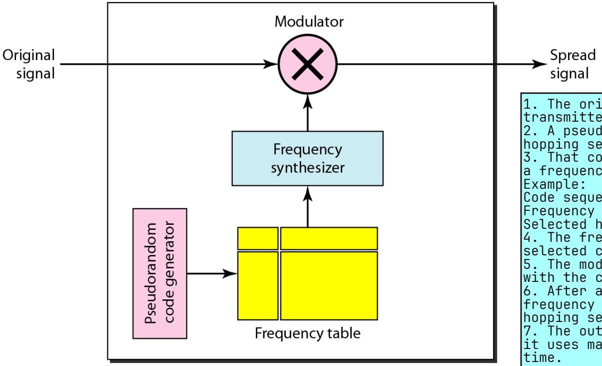


Figure 6.28: FHSS

Figure: In FHSS, a signal hops among multiple carrier frequencies over time according to a hopping sequence.



1. The original signal/data enters the transmitter.
2. A pseudorandom code generator creates a hopping sequence.
3. That code sequence selects frequencies from a frequency table.

Example:
Code sequence: 3, 1, 4, 2
Frequency table: f1, f2, f3, f4
Selected hops: f3 → f1 → f4 → f2

4. The frequency synthesizer generates the selected carrier frequency.
5. The modulator combines the original signal with the current carrier frequency.
6. After a short time interval, the carrier frequency changes to the next one in the hopping sequence.
7. The output becomes a spread signal because it uses many different carrier frequencies over time.

The receiver must use the same pseudorandom code and frequency table so it can hop in the same pattern and recover the signal.

Figure 6.29: Frequency Selection in FHSS

Figure: A pseudo-random control process selects the next carrier frequency for each hop interval.

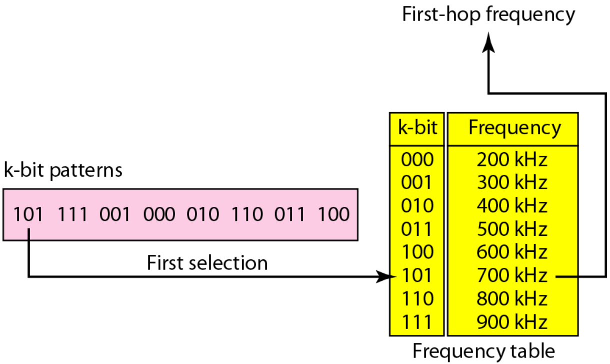


Figure 6.30: FHSS Cycles

Figure: Hopping patterns repeat in cycles; transmitter and receiver must stay sequence-synchronized.

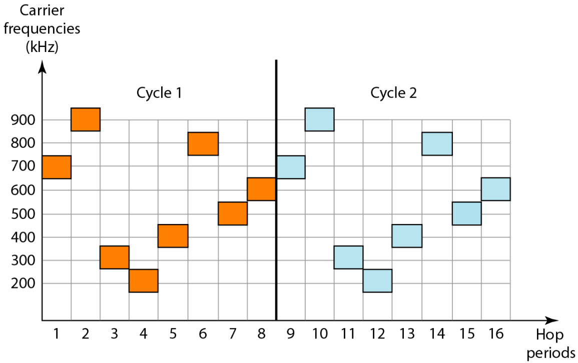


Figure 6.31: Bandwidth Sharing in FHSS

Figure: With M hopping frequencies, multiple channels can share the spread band through coordinated hopping patterns.

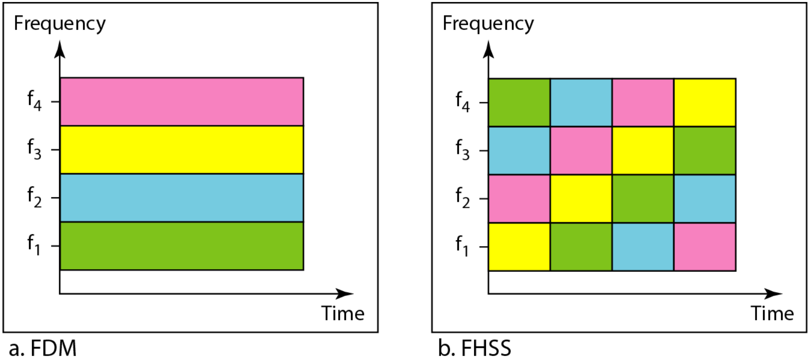
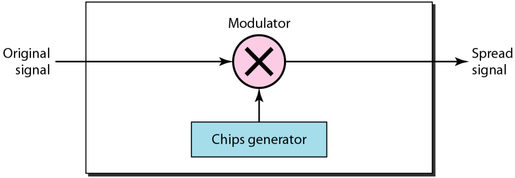


Figure 6.32: DSSS

- Each data bit is replaced by n faster chips.
- Chips come from a spreading code.
- Chip rate is n times the data-bit rate.



Unlike FHSS, where the carrier frequency jumps from one frequency to another, DSSS keeps using the same general carrier band, but it spreads the data by mixing each bit with a much faster code sequence.
Each data bit is replaced by n faster chips.
That means one original bit is not sent as just one bit-duration signal. Instead, it is expanded into several smaller pieces called chips.
Example:

Original data signal: 1 0

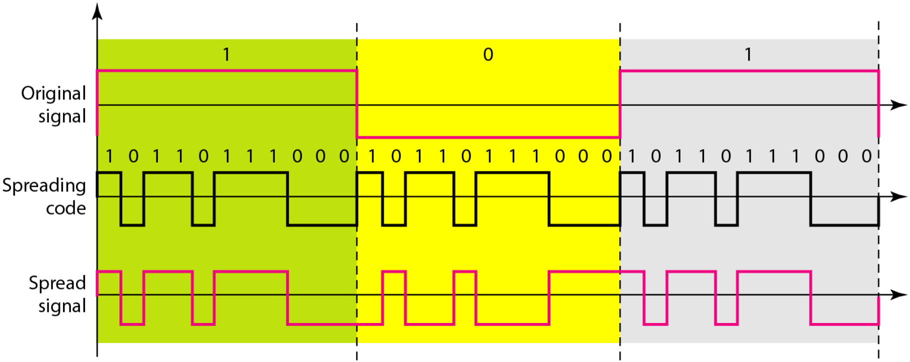
Spreading code with n = 8:
1: 10110100
0: 01001011

Transmitted chips:
10110100 01001011

So instead of sending 2 bits we send 16 chips representing the 2 bits if the original data rate is 'b' kbps and each bit is replaced by 'n' chips, then the chip rate becomes: nb kbps

Figure 6.33: DSSS Example

Figure: Bit-by-chip spreading expands occupied bandwidth, enabling later despreading with the correct code.



Summary (1)

- Bandwidth utilization means using available bandwidth to achieve specific goals.
- **Multiplexing** improves efficiency; **spreading** improves robustness/privacy against jamming and interception.
- Multiplexing enables simultaneous transmission of multiple signals over one data link.
- Three core multiplexing families are FDM, WDM, and TDM.
- WDM applies analog multiplexing ideas to optical wavelengths in fiber systems.

Summary (2)

- TDM is a digital process that shares one high-rate link among several low-rate streams in time.
- Synchronous TDM uses fixed slot positions even when a source is idle.
- Statistical TDM assigns slots dynamically and improves efficiency for bursty traffic.
- Spread spectrum is especially useful in wireless settings where coexistence, anti-jamming, and anti-interception matter.

Summary (3)

- FHSS spreads by hopping carriers among many frequencies over time.
- DSSS spreads by replacing each data bit with a chip sequence.
- Both techniques trade extra bandwidth for improved robustness and sharing behavior in hostile or crowded spectrum.